

Thermal Management Deep Dive

θ_{JA} • Heat spreading • Thermal vias • Heatsinks • Active cooling

Heat kills reliability — every +10 °C halves life

Audience: Intermediate Engineering Students

Conduction • Convection • Radiation • PCB tactics • Validation

Why Thermal Matters

Heat is the #1 reliability killer

Lifetime

Arrhenius — every +10 °C halves component life.

Performance

MCUs throttle at T_{jmax}
LDOs go into shutdown.

Spec

Junction temp T_j specified
in every IC datasheet.

Cost

Smaller heatsink = lower cost
Good layout = no heatsink.

Compliance

Touch-temperature limits
for safety (UL, IEC).

Field returns

Electrolytic cap dry-out
→ #1 RMA cause.

Heat Transfer Basics

Three modes — all happen on a PCB

- ▶ Conduction: through solid materials (Cu, FR-4, IC package)
 - ▶ $Q = k \times A \times \Delta T / L$ (k = thermal conductivity)
 - ▶ Cu: 400 W/mK; FR-4: 0.3 W/mK; AlN ceramic: 180 W/mK
- ▶ Convection: heat carried away by moving air or liquid
 - ▶ Natural convection: ~5 W/m²K • Forced air: 25-100 W/m²K • Liquid: 500-5000
- ▶ Radiation: emitted as IR — significant at high temperatures (> 80 °C)
 - ▶ $Q = \sigma \times \epsilon \times A \times (T^4 - T_{\text{amb}}^4)$
- ▶ PCB design optimizes CONDUCTION (heat from IC into copper into board) + CONVECTION (heat off the board)

θ (Theta) Parameters

How IC datasheets quantify thermal performance

Parameter	Meaning	Typical (SOIC-8)	Typical (QFN-32 with thermal pad)
θJA	Junction-to-Ambient	150 °C/W	30-40 °C/W
θJC	Junction-to-Case (top)	30 °C/W	5-10 °C/W
θJB	Junction-to-Board	40 °C/W	10-15 °C/W
ψJT	Junction-to-Top (for measurement)	5-10 °C/W	2-5 °C/W
TJmax	Max allowed junction temp	150 °C	125-150 °C
PDmax	Max power dissipation	(TJmax-Ta)/θJA	Same calc, but with θJA

Worked example

TJmax = 125 °C, Ta = 60 °C, θJA = 40 °C/W. PDmax = (125-60)/40 = 1.6 W. If your IC draws 2 W → you need heatsink OR more copper area.

Thermal Pad + Vias

How modern QFN / DFN packages dump heat

- ▶ Modern QFN / DFN / PowerSOIC packages have a large CENTER PAD
- ▶ Pad is electrically GND (or 'no-connect' depending on part)
- ▶ Pad mates to a matching copper area on the PCB
- ▶ Underneath: stitching VIAS to the bottom-side copper pour or inner plane
- ▶ Vias provide thermal path from IC die → bottom-side copper → ambient air
- ▶ Typical pattern: 4×4 to 6×6 grid of 0.3 mm vias under thermal pad
- ▶ Via-in-pad option: filled & plated for premium thermal performance
- ▶ Solder mask opening: full pad area (no mask in the via region)

Copper Pour as Heat-Spreader

Bigger pour = more heat dissipation

- ▶ Cu conducts heat 1300× better than FR-4
- ▶ Pour around hot ICs spreads heat across the board area
- ▶ Typical effect: doubling pour area halves θ_{JA}
- ▶ Rule of thumb: 1 cm² of 1 oz pour $\approx$ 30-50 °C/W to ambient
- ▶ Multiple inner layer GND pours connected via thermal pad vias → additive
- ▶ Avoid thermal isolation by trace cuts — keep pour CONNECTED
- ▶ DRC: 'thermal spoke' relief for through-hole pins (prevents pin from sucking heat from pour during soldering)
- ▶ Color of solder mask affects radiation slightly (black > green > white)

Heatsinks

When copper pour isn't enough

Clip-on heatsink

Spring clip onto IC body.
No screw needed.
3-5 °C/W typical.

Screw-mounted

Two screws + thermal pad.
Better thermal contact.
1-3 °C/W.

Bonded heatsink

Thermal adhesive on IC top.
No mechanical hardware.
4-6 °C/W.

Heat-pipe-extended

Heat pipe to remote heatsink.
Used in laptops, dense designs.
2-4 °C/W.

Liquid cooling

Water block on IC.
High-performance servers, GPUs.
0.1-1 °C/W.

PCB-as-heatsink

Thick copper PCB acts as heatsink.
No separate hardware.
5-10 °C/W.

Forced Air Cooling

When passive isn't enough

- ▶ Natural convection: $\sim 5\text{-}10\text{ W/m}^2\text{K}$ — limited
- ▶ Forced air (fan): $25\text{-}100\text{ W/m}^2\text{K}$ — 5-10× improvement
- ▶ Fan selection: airflow (CFM) + back-pressure (mm-H₂O) + noise (dBA)
- ▶ Place IC in airflow path — don't shield with tall components
- ▶ Cross-board airflow > top-down for densely populated boards
- ▶ Fan speed control via PWM + thermistor — quiet at low load, ramps with temp
- ▶ Failure mode: fan dies → IC overheats → product fails
- ▶ Always design for natural convection as backup; fan is bonus

Thermal Interface Materials (TIM)

Between IC and heatsink — must fill air gaps

TIM type	Thermal conductivity	Use	Cost
Thermal grease (paste)	3-12 W/mK	Most common, repaintable	Cheap
Phase-change pad	3-5 W/mK	Solid at RT, liquefies hot	Mid
Thermal pad (silicone)	1-5 W/mK	Pre-cut, easy assembly	Cheap
Graphene / metallic pad	10-100 W/mK	High-end CPUs / GPUs	Expensive
Liquid metal (gallium)	70-80 W/mK	Extreme overclocking	Expensive + risky
Adhesive epoxy	1-3 W/mK	Permanent bond	Mid

Why TIM matters

Air gaps between IC top and heatsink (microscopic, even on flat surfaces) have very low thermal conductivity (~0.025 W/mK). TIM fills the gaps and dramatically reduces effective $\theta_{\text{case-to-heatsink}}$.

Component Selection for Low Heat

Best to AVOID the problem

- ▶ Pick switching regulator over LDO when ΔV is large (LDO dissipates the difference as heat)
- ▶ Pick MOSFET with low $R_{DS(on)}$ — lower conduction losses
- ▶ Pick lower-frequency switching when possible (fewer transitions, less heat)
- ▶ Pick higher-efficiency IC variants (newer = usually more efficient)
- ▶ Pick LARGER package — more thermal area at same cost
- ▶ Avoid mixing high-power on multi-channel ICs (e.g., 4-channel amp at full power)
- ▶ Distribute heat across multiple ICs rather than concentrating in one
- ▶ Component derating: 70% of rated power max → halves dissipation

Layout Practices for Thermal

What separates thermal-aware designs from naïve ones

- ▶ Hot ICs SPACED OUT — don't cluster regulators + MCUs + power amps
- ▶ Heat-sensitive parts (electrolytic caps, sensors) AWAY from hot ICs
- ▶ Thermal pad + via stitching MANDATORY for power-dissipating QFN/DFN
- ▶ Copper pour CONNECTED across multiple ground regions for spreading
- ▶ Use 2 oz outer Cu for hot regions (better heat conduction)
- ▶ Thermal vias around connectors that carry high current (USB-C, power)
- ▶ Avoid heat-sensitive cooled chips next to switching regulators (thermal coupling)
- ▶ Plan placement based on TYPICAL OPERATING POINT, not idle

Thermal Simulation Tools

Predict before you fab

Ansys Icepak

Full CFD + thermal
Industry standard
Licensed

FloEFD / FloTHERM

Mentor's CFD tool
EDA-integrated

Cadence Sigrity Celsius

Cadence Allegro
Licensed

Online vendor calcs

TI, ADI, ST app notes
Free, IC-specific

KiCad

No native thermal sim
Use external tools

Spreadsheet

Hand calcs with θ_{JA} , θ_{JC}
Good for budgeting

Bench Validation

How to confirm your thermal design works

- ▶ Thermal camera (₹15-50k FLIR / Seek) — see hot spots in real time
- ▶ Thermocouple + DMM at IC junction / case — quantitative measurement
- ▶ Inrush + steady-state — measure during both
- ▶ Worst-case ambient: stress test at 40-60 °C in a thermal chamber
- ▶ Long-duration test (24 hrs) — find slow drifts (e.g., cap warming up)
- ▶ Compare against vendor's calculated PDmax
- ▶ TJmax: derate by 15-20 °C below datasheet max for reliability margin
- ▶ Document: thermal profile per major IC + worst-case junction temp

Common Thermal Mistakes

What field returns teach

- ▶ Using TQFP / SOIC where QFN with thermal pad would be appropriate
- ▶ Thermal pad designed in but no stitching vias to bottom copper
- ▶ Hot IC placed next to electrolytic cap → cap lifetime ruined
- ▶ 1 oz Cu when 2 oz is needed for thermal spreading
- ▶ Using θ_{JA} without copper area context (datasheet θ_{JA} assumes 1 oz, large pour)
- ▶ Designing at room temperature (25 °C) — but product runs at 50-60 °C ambient
- ▶ Heatsink with poor thermal paste / no paste → high case-to-heatsink resistance
- ▶ Forgetting fan dies → no plan B → product cooks

Key Takeaways

Key takeaways from this lecture

Arrhenius rule

Every +10 °C halves life.

θ_{JA} is the spec

Use it to budget power.

Thermal pad + vias

Mandatory for power QFN.

Copper = heatsink

Pour spreads, vias conduct.

Pick cool components

LDO → buck where possible.

Validate with camera

Real measurements only.

Questions & Discussion

Ask anything — concepts, projects, sourcing, career.

References & Further Learning

- ▶ Henry Ott — EMC book has thermal sections
- ▶ Texas Instruments — 'Thermal Design Guide' app notes
- ▶ Analog Devices — MT-093 'Thermal Considerations'
- ▶ Ansys Icepak / FloTHERM — commercial CFD tools
- ▶ FLIR thermal-imaging tutorials
- ▶ Robert Feranec YouTube — thermal management in PCBs
- ▶ IPC-2152 — current-carrying capacity (related to heat)
- ▶ Local fab heavy-Cu offerings for power thermal designs

Thank you • Build something this week.